

Solving Exponential and Logarithmic Equations

Answer Key

Algebra 2

Quick Answers

- | | |
|------------|--|
| 1. 4 | 7. 4 |
| 2. 13 | 8. 20.84 days |
| 3. 40500 | 9. 2 |
| 4. 4.32 | 10. (a) the exact solution = $\frac{8}{3}$, (b) the solution as a decimal = 2.67, — |
| 5. 2 | |
| 6. 3581.70 | |
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What is verified

11 of 12 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSF-BF.B.5 – *problems 2, 7, 10*

HSF-LE.A.4 – *problems 1, 3, 4, 5, 6, 8, 9*

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

7. *If they got 4, -2: solved the quadratic and kept $x = -2$, which makes both logarithm inputs negative*

1. Solve $3^x = 81$.

Solution: 81 is a power of 3: $81 = 3^4$. With equal bases the exponents must be equal.

$x = 4$

2. Solve $\log_4(x + 3) = 2$.

Solution: Rewrite in exponential form: $4^2 = x + 3$, so $16 = x + 3$. The input at $x = 13$ is $16 > 0$, so the answer stands.

$x = 13$

3. $N = 500 \cdot 3^t$ bacteria; find N at $t = 4$ hours.

Solution: Substitute $t = 4$ and evaluate the power first: $3^4 = 81$, so $N = 500 \times 81$.

40500

4. Solve $2^x = 20$ to two decimal places.

Solution: 20 is not a power of 2, so take logarithms of both sides and bring the exponent down with the power rule:

$$x \ln 2 = \ln 20, \quad x = \frac{\ln 20}{\ln 2} = \frac{2.99573\dots}{0.69314\dots} = 4.32192\dots$$

$x \approx 4.32$

5. Solve $5^{2x-1} = 125$.

Solution: $125 = 5^3$, so $5^{2x-1} = 5^3$ and the exponents match: $2x - 1 = 3$, giving $2x = 4$.

$x = 2$

6. $A = 2000(1.06)^t$ dollars; find A at $t = 10$ years.

Solution: Evaluate the growth factor first, then multiply:

$$1.06^{10} = 1.7908476\dots, \quad A = 2000 \times 1.7908476\dots = 3581.6953\dots$$

To the nearest cent, in dollars,

3581.70

7. Solve $\log_2 x + \log_2(x - 2) = 3$.

Solution: Condense with the product rule, then switch to exponential form:

$$\log_2(x(x - 2)) = 3$$

$$x(x - 2) = 2^3 = 8$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0.$$

The candidates are $x = 4$ and $x = -2$. At $x = -2$ both inputs, x and $x - 2$, are negative, so neither logarithm exists — that candidate is discarded. At $x = 4$: $\log_2 4 + \log_2 2 = 2 + 1 = 3$
 $\checkmark$

$x = 4$

8. $A = 100(0.5)^{t/12}$ mg; find t when $A = 30$ mg.

Solution: Divide by 100 first so the power stands alone, then take logarithms:

$$\begin{aligned}(0.5)^{t/12} &= 0.3 \\ \frac{t}{12} \ln 0.5 &= \ln 0.3 \\ \frac{t}{12} &= \frac{\ln 0.3}{\ln 0.5} = 1.736965 \dots \\ t &= 20.8435 \dots\end{aligned}$$

Rounded to two decimal places,

20.84 days

9. Solve $9^{x+1} = 27^x$.

Solution: Both 9 and 27 are powers of 3, so rewrite everything in base 3:

$$(3^2)^{x+1} = (3^3)^x \implies 3^{2x+2} = 3^{3x}.$$

Equal bases give $2x + 2 = 3x$. Check: $9^3 = 729 = 27^2$ ✓

$x = 2$

10. Challenge: $\log_5(2x + 3) - \log_5(x - 1) = 1$. (a) Exact x . (b) Decimal. (c) The domain condition.

Solution (a): Condense with the quotient rule, then switch form:

$$\begin{aligned}\log_5 \frac{2x+3}{x-1} &= 1 \\ \frac{2x+3}{x-1} &= 5^1 = 5 \\ 2x+3 &= 5x-5 \\ 3x &= 8.\end{aligned}$$

$x = \frac{8}{3}$

Solution (b): $\frac{8}{3} = 2.666 \dots$

2.67

Solution (c): *Instructor-judged.* Both inputs must be positive: $2x+3 > 0$ gives $x > -\frac{3}{2}$, and $x-1 > 0$ gives $x > 1$. Together the domain is $x > 1$. Since $\frac{8}{3} \approx 2.67 > 1$, the solution is legitimate. The condensing step is only an identity on that domain, so a candidate outside it would satisfy the equation condensing produced while making the original expression undefined — it would have to be discarded. Full credit states $x > 1$ and confirms the solution meets it.